

Project Design Phase-II Technology Stack (Architecture & Stack)

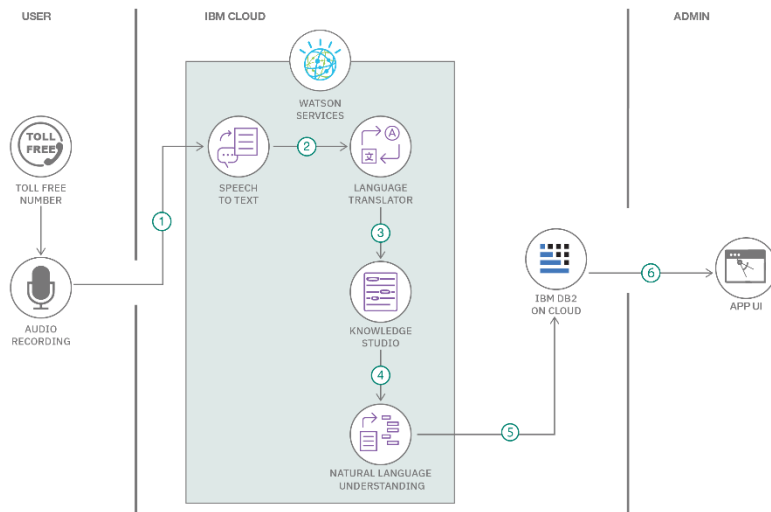
Date	31 January 3035
Team ID	
Project Name	House Rent App
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1. 1	Frontend	React.js, Tailwind CSS, Redux for state management	React JS, Tailwind CSS
2. 2	Backend	Node.js, Express.js for REST API	Node.js, Express.js
3. 3	Database	MongoDB (NoSQL) for flexible property schema	MongoDB (NoSQL)
4. 4	Authentication	JWT (JSON Web Tokens)	MongoDB Atlas
5. 5	Hosting	AWS / Vercel / Render	Cloudinary
6. 6	Version Control	Git & GitHub	Render / Vercel
7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	React, Express, Node.js
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	JWT, bcrypt

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	REST API
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Cloud Hosting (Vercel/Render)
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Image optimization via Cloudinary

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>